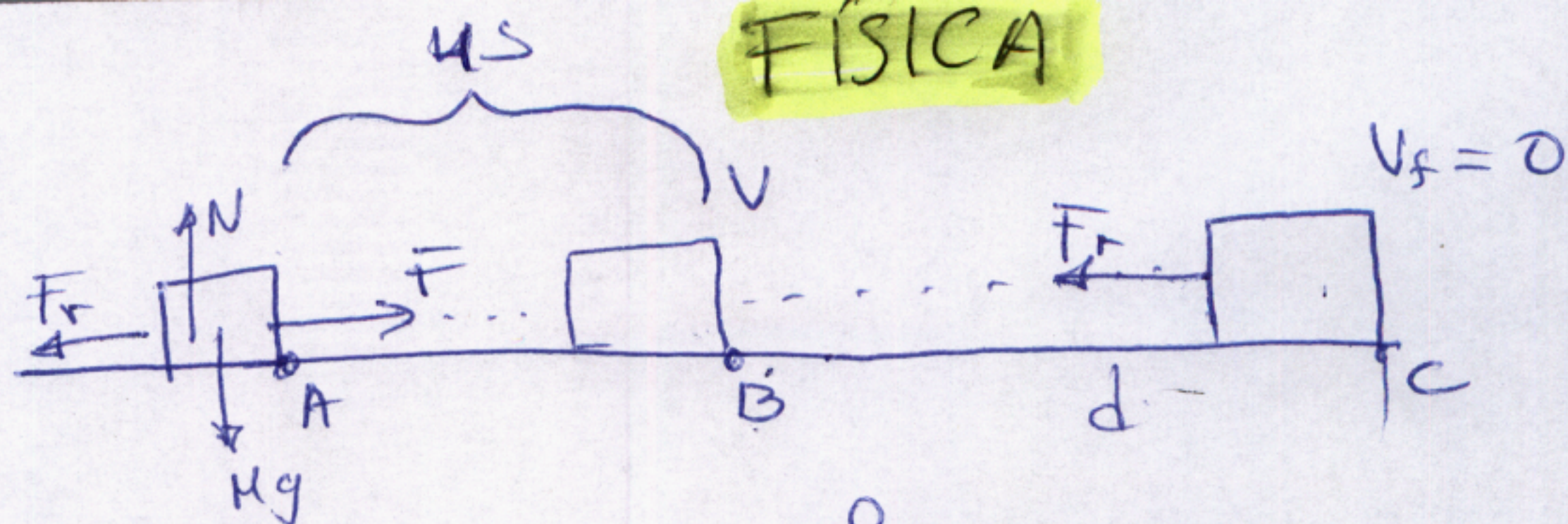


5) $M = 1 \text{ kg}$

$F = 3 \text{ N}$
 $t = 4 \text{ s}$
 $\mu = 0,2$

$t_d = ?$



$$F - F_r = M \cdot a$$

$$F - N \cdot \mu = M \cdot a$$

$$F - Mg \cdot \mu = M \cdot a$$

$$a = \frac{F - M \cdot g \cdot \mu}{M}$$

$$a = 1 \text{ m/s}^2$$

$$V = v_0 + a \cdot t$$

$$V = 1 \text{ m/s}^2 \cdot 4 \text{ s}$$

$$V = 4 \text{ m/s}$$

Trauro B-C.

$$\frac{1}{2} M \cdot V^2 = Q_{Bc}$$

$$\frac{1}{2} M V^2 = F_r \cdot d$$

$$\frac{1}{2} M \cdot V^2 = M \cdot g \cdot \mu \cdot d$$

$$d = \frac{M \cdot V^2}{2g \cdot \mu} = \frac{(4 \text{ m/s})^2}{2 \times 10 \text{ m/s}^2 \times 0,2} = 4 \text{ m}$$

$$d = d_0 + v \cdot t + \frac{1}{2} a t^2$$

$$v_f^2 = v_0^2 + 2a \cdot d$$

$$d = d_0 + v \cdot t + \frac{1}{2} a t^2$$

$$v_f^2 = v_0^2 + 2a \cdot d$$

$$t_d = \frac{v}{a}$$

$$d = \frac{v \cdot v}{a} - \frac{1}{2} a \cdot \left(\frac{v}{a}\right)^2$$

$$d = \frac{4^2}{a} - \frac{4^2}{2a}$$

$$\Rightarrow t_d = \frac{v}{a} = \frac{4 \text{ m/s}}{2 \text{ m/s}^2} = 2 \text{ s}$$

$$4 = \frac{1}{2} \cdot \frac{4^2}{a}$$

$$a = \frac{16}{8} = 2 \text{ m/s}^2$$

$$t_d = 2 \text{ [s]}$$

SOLUCIONARIO SEGUNDO PARCIAL

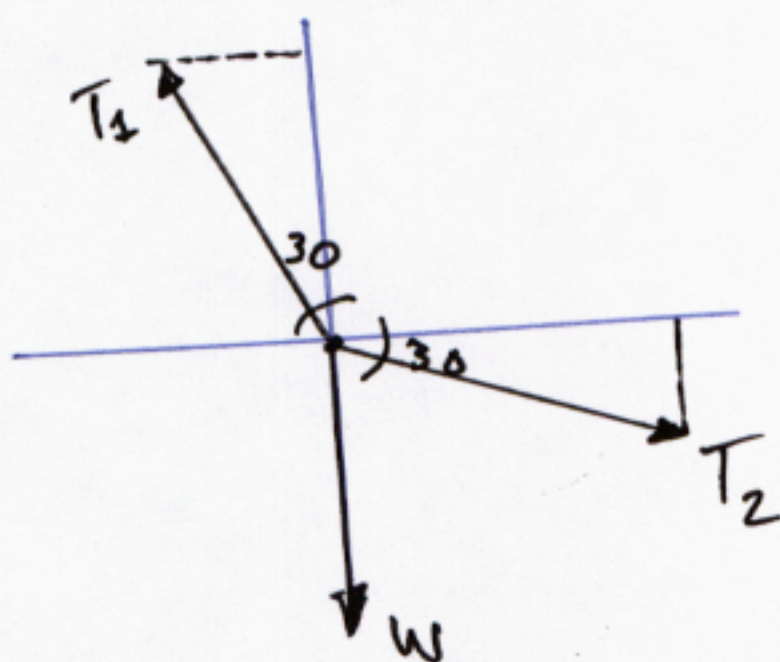
ASIGNATURA:

FECHA:

PÁGINAS:

..... de

F.2



$$\sum F_x = 0$$

$$T_2 \cos 30^\circ - T_1 \sin 30^\circ = 0 \quad (1)$$

$$\sum F_y = 0$$

$$T_1 \cos 30^\circ - W - T_2 \sin 30^\circ = 0 \quad (2)$$

$$\text{de (1)} \quad T_1 = \frac{T_2 \cos 30^\circ}{\sin 30^\circ}$$

$$\text{en (2)} \quad \frac{T_2 \cos 30^\circ \cdot \cos 30^\circ}{\sin 30^\circ} - W - T_2 \sin 30^\circ = 0$$

$$\frac{3}{2} T_2 - 240 - T_2 \cdot \frac{1}{2} = 0$$

$$\boxed{T_2 = 240 \text{ N}}$$

de (3)

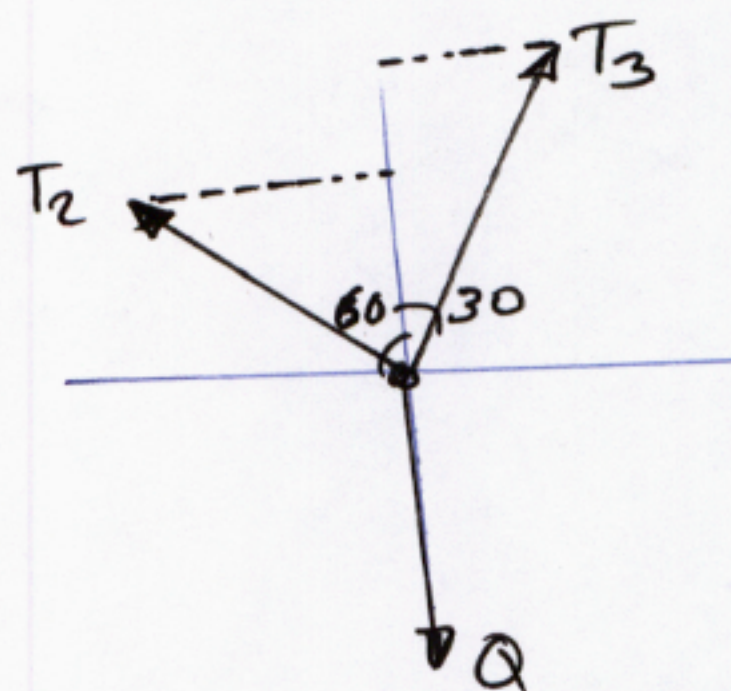
$$T_3 = \frac{T_2 \sin 60^\circ}{\sin 30^\circ} = \sqrt{3} \cdot 240 \text{ N}$$

$T_3 \text{ en (4)}$

$$Q = T_3 \cos 30^\circ + T_2 \cos 60^\circ$$

$$Q = \sqrt{3} \cdot 240 \cdot \frac{\sqrt{3}}{2} + 240 \cdot \frac{1}{2}$$

$$\boxed{Q = 480 \text{ N}}$$



$$\sum F_x = 0$$

$$T_3 \sin 30^\circ - T_2 \sin 60^\circ = 0 \quad (3)$$

$$\sum F_y = 0$$

$$T_3 \cos 30^\circ + T_2 \cos 60^\circ - Q = 0 \quad (4)$$

⑩ F.3.

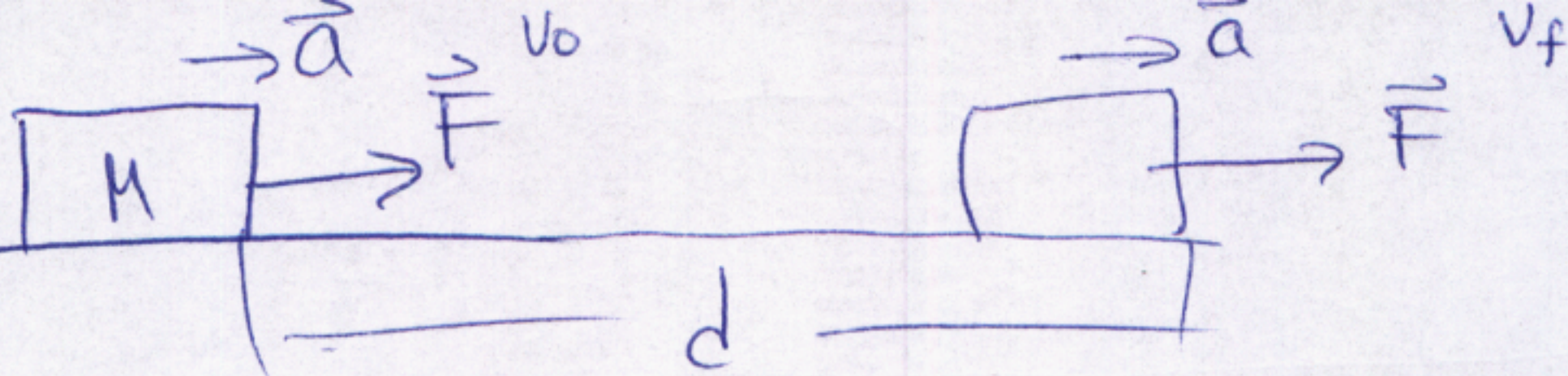
$$M = 500 \text{ kg}$$

$$v_0 = 0$$

$$v_f = 20 \text{ m/s}$$

$$d = 100 \text{ m}$$

$$F = ?$$



~~$$d = v_0 t + \frac{1}{2} a t^2$$~~

$$v_f^2 = v_0^2 + 2a \cdot d$$

$$a = \frac{v_f^2}{2d} = \frac{(20 \text{ m/s})^2}{2 \times 100 \text{ m}} = 2 \text{ m/s}^2$$

$$F = M \cdot a$$

$$F = 500 \text{ kg} \cdot 2 \text{ m/s}^2 = 1000 \text{ N}$$

$$\boxed{F = 1000 \text{ N}}$$

SOLUCIONARIO SEGUNDO PARCIAL

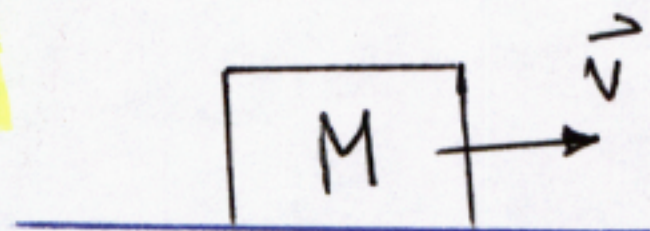
ASIGNATURA:

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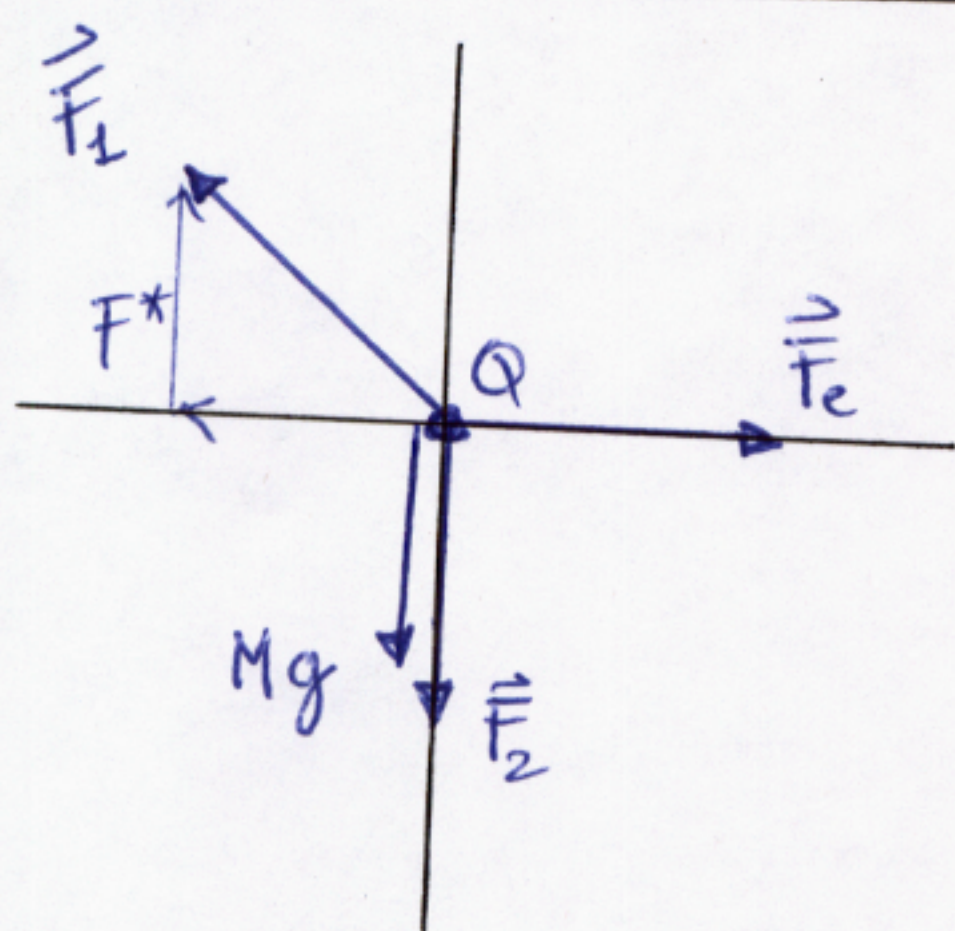
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F.4.



$$E_k = \frac{1}{2} M \cdot v^2 \Rightarrow v = \sqrt{\frac{2 E_k}{M}} = \sqrt{\frac{2 \times 3200}{4}} = 40 \text{ m/s}$$

F.5

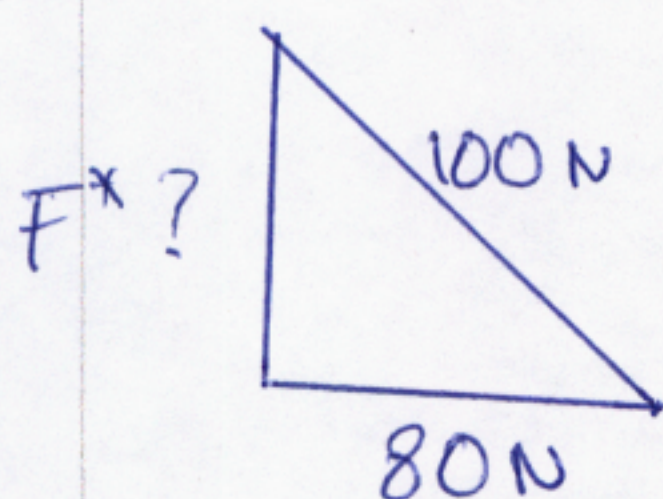


$$F_e = Q \cdot E = 20 \times 10^{-3} \text{ C} \cdot 4 \times 10^3 \frac{\text{N}}{\text{C}}$$

$$F_e = 80 \text{ N}$$

$$F_1 = K_1 \cdot \Delta X_1 = 25 \frac{\text{N}}{\text{cm}} \cdot 4 \text{ cm}$$

$$F_1 = 100 \text{ N}$$



$$100^2 = 80^2 + F^{*2}$$

$$F^* = \sqrt{100^2 - 80^2}$$

$$F^* = 60 \text{ N}$$

$$\Sigma F_y = 0$$

$$F^* - Mg - F_2 = 0$$

$$F^* - Mg - K_2 \cdot \Delta X_2 = 0$$

$$\Delta X_2 = \frac{F^* - Mg}{K_2} = \frac{60 \text{ N} - 2 \text{ kg} \cdot 10 \text{ m/s}^2}{2000 \text{ N/m}}$$

$$\Delta X_2 = 0,02 \text{ m} \quad \& \quad \Delta X_2 = 2 \text{ cm}$$